



Software Requirements Specification

e-Pili Bomba Kunak — Version 2 (V2)

Spesifikasi Keperluan Perisian · Sistem Rekod Pili Bomba

Organisation / Organisasi: Bahagian Bomba dan Penyelamat (BBP) Kunak, JBPM Sabah

System / Sistem: epilibomba.com · Fire-Hydrant Inspection & Record Application

Document standard: IEEE Std 830-1998 (adapted)

Status: LIVE in production · Single district (Kunak)

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1.0	2026-08-18	PM / Eng	Initial baseline SRS capturing the live V2 system, technology rationale, and forward roadmap.

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1. Introduction / Pengenalan

1.1 Purpose / Tujuan

This Software Requirements Specification (SRS) defines the functional and non-functional requirements, the technology stack and its justification, and the forward roadmap for **e-Pili Bomba Kunak Version 2 (V2)** — a mobile-first web application used by BBP Kunak officers to locate fire hydrants (pili bomba), record inspections, and maintain the legally-controlled inspection record card (Kad Rekod Pili Bomba) under MS ISO 9001:2015 procedure PS-8.

Tujuan: Dokumen ini menetapkan keperluan berfungsi dan bukan-berfungsi, teknologi yang digunakan beserta justifikasi, dan pelan masa hadapan bagi sistem e-Pili Bomba Kunak Versi 2 — aplikasi web mudah alih untuk pegawai BBP Kunak mengesan pili bomba, merekod pemeriksaan, dan menyelenggara Kad Rekod Pili Bomba di bawah prosedur MS ISO 9001:2015.

1.2 Scope / Skop

The system provides: an interactive hydrant map, a two-page inspection record card, digital signatures, an inspection dashboard, a shared inspection schedule (jadual), role-based access, and full offline capability. It is live at epilibomba.com, currently serving **one district (Kunak, Sabah)**, and is architected to expand to additional districts and, in future, state/national scale without re-writing the map engine. The user-facing language is Bahasa Malaysia.

Skop: Sistem menyediakan peta pili bomba interaktif, kad rekod pemeriksaan dua muka surat, tandatangan digital, papan pemuka pemeriksaan, jadual pemeriksaan dikongsi, kawalan akses mengikut peranan, dan keupayaan luar talian penuh. Kini melayani satu daerah (Kunak) dan direka untuk berkembang ke daerah lain tanpa menulis semula enjin peta.

1.3 Definitions, Acronyms & Abbreviations / Definisi & Singkatan

Term	Meaning
Pili Bomba	Fire hydrant.
Kad Rekod	The inspection record card — a controlled MS ISO document (2 pages).
Awam / Swasta	Public (government-maintained) / Private hydrant.
Jadual	Shared inspection schedule.
RLS	Row-Level Security (PostgreSQL access control).
SPA	Single-Page Application.
CSP	Content Security Policy.
JBPM	Jabatan Bomba dan Penyelamat Malaysia.
V1 / V2	Version 1 (single static <code>index.html</code> , frozen rollback) / Version 2 (Vue build, current).

1.4 References / Rujukan

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications.
- Project documents: PRD.md , FULL-ARCHITECTURE.md , KAD-REKOD.md , CUTOVER.md , STAGING.md , RESTORE.md , V2-ROADMAP.md .
- MS ISO 9001:2015 — procedure PS-8 Pili Bomba (record-card layout & retention).

1.5 Overview / Gambaran

Section 2 describes the product and its users. Section 3 justifies each technology choice. Section 4 lists functional and non-functional requirements. Section 5 sets out the forward roadmap, including the organisational actions (items 2–9) required before the system scales beyond one district.

2. Overall Description / Perihal Keseluruhan

2.1 Product Perspective / Perspektif Produk

A self-contained web SPA delivered as a static bundle from a CDN, talking to a managed PostgreSQL backend (Supabase) over HTTPS. There is no custom application server: the browser is the client, and all business rules that matter for integrity are enforced in the database (RLS policies + triggers). V1 (a single static `index.html`) remains deployed and untouched as the one-minute rollback.

2.2 Product Functions / Fungsi Produk

- Interactive hydrant map with clustering, status filters, zone filters, and place search.
- Two-page inspection record card (Kad Rekod) with automatic card growth and permanent signed rows.
- Digital signature capture, reusable profile signature, and print-accurate signature rendering.
- Inspection dashboard (derived — stores no numbers of its own) with a hand-built 3D donut.
- Shared inspection schedule (jadual), admin-writable.
- Role-based access (viewer / admin) enforced by RLS.
- Full offline operation with automatic sync and conflict recovery.

2.3 User Classes & Characteristics / Kelas Pengguna

Class	Rights	Notes
Viewer	Read-only across all data (mutual aid).	Default for new accounts.
Admin	Read all; write (edit records, sign, add hydrants, edit jadual).	Writes are scoped to the officer's own district.
Upper-admin (planned)	District manager — provisions accounts within their district.	Designed, not built (Section 5).

Kelas pengguna: *Viewer (baca sahaja), Admin (baca & tulis dalam daerah sendiri), dan Upper-admin/Pengurus Daerah (dirancang) yang menguruskan akaun dalam daerahnya.*

2.4 Operating Environment / Persekitaran Operasi

- **Client:** modern mobile browsers (Android/iOS), used one-handed in the field, sometimes in gloves, often on weak/intermittent signal.
- **Delivery:** static bundle on Cloudflare Pages (global CDN + edge).
- **Backend:** Supabase (PostgreSQL 15, Auth, Storage) — project `isxfhocfkjamjchmicwq`.
- **Map tiles:** OpenStreetMap via Leaflet.

2.5 Design & Implementation Constraints / Kekangan

- **Legal record:** a signed Kad Rekod row is **permanent** — never editable, clearable, or deletable — enforced in RLS *and* an independent trigger *and* the client.

- **Security:** `script-src 'self'` — no third-party scripts; all vendor libraries self-hosted. Audit identity is taken from the database JWT, never the request body.
- **Print fidelity:** the record card is tuned to the millimetre for A4/Letter; print CSS order is load-bearing.
- **Reads global, writes district-scoped** (mutual aid between stations).
- **No data loss with no signal** — offline work is parked and synced.

2.6 Assumptions & Dependencies / Andaian & Kebergantungan

- Supabase remains available and within plan limits (free tier today — see Section 5, item 7).
- Cloudflare Pages remains the host; the DNS custom domain is the rollback switch.
- Each officer has an individual login (signatures attribute who signed).

3. Technology Stack & Rationale / Teknologi & Justifikasi

Each technology below is stated with **what it is**, **the version in use**, and **why it was chosen** for this specific government field application.

3.1 Vite (build tool — v8.2.1)

What: a fast front-end build tool and bundler that compiles Vue single-file components into an optimised static bundle.

Why chosen: V2 needs a build step (V1 had none), and Vite gives a predictable, single-artifact output that suits the strict security posture: one external CSS file (`cssCodeSplit:false`) so the millimetre-tuned print stylesheet arrives in a known order; no inline scripts, so the CSP can stay `script-src 'self'` with no `'unsafe-inline' / 'unsafe-eval'` ; and nothing silently inlined (`assetsInlineLimit:0`). It is fast enough that the whole app builds in ~1 second, keeping the release gate cheap.

Vite: alat binaan pantas yang menghasilkan satu bundle statik. Dipilih kerana menyokong dasar keselamatan ketat (tiada skrip luaran) dan susunan CSS cetakan yang tepat untuk Kad Rekod.

3.2 Vue 3 (UI framework — v3.5.41)

What: a progressive, component-based JavaScript framework for building reactive user interfaces.

Why chosen: V1 was one 3,600-line file that had become hard to change safely. Vue 3's component model breaks the app into testable, single-responsibility pieces (map, card, dashboard, schedule) while keeping a gentle learning curve for a small team. The **runtime-only** build is used deliberately — no template compiler in the browser — which removes `new Function/eval` and lets the CSP stay strict. Vue's reactivity drives the live dashboard and offline-sync indicators without manual DOM wiring.

Vue 3: kerangka antara muka berasaskan komponen. Dipilih untuk memecahkan aplikasi kepada bahagian yang boleh diuji dan diselenggara, sambil mengekalkan keselamatan (tiada penyusun templat dalam pelayar).

3.3 Pinia (state management — v3.0.3)

What: the official state-management library for Vue 3 — centralised, reactive stores.

Why chosen: the app has genuinely shared state (the signed-in session, the register, scope filters, the offline queue) read by multiple components. Pinia gives one source of truth for that state, and — critically for this app — it enabled a clean split between **pure logic modules** (`*-logic.js` , no framework, no browser) and thin Pinia stores. That split is what lets the most dangerous code (the offline sync decision table) be unit-tested in isolation against the V1 baseline.

Pinia: pengurusan keadaan rasmi untuk Vue. Dipilih kerana membolehkan satu sumber kebenaran untuk data dikongsi, dan memisahkan logik tulen (boleh diuji) daripada antara muka.

3.4 SQL / PostgreSQL (database engine — PostgreSQL 15, via Supabase)

What: a mature, standards-compliant relational database; the data is modelled in SQL tables with constraints, Row-Level Security policies, and triggers.

Why chosen: the Kad Rekod is a **legal record**, so integrity cannot live in the client. PostgreSQL enforces the rules where they cannot be bypassed: **RLS** decides who may read/write; a **trigger** makes signed rows permanent independently of the app; the audit column is stamped from the login token, not the request body. SQL also scales trivially to national volume (~100k rows) with ordinary indexes, and its DR scripts (sql/) are the exact artefact a recovery re-applies.

SQL / PostgreSQL: pangkalan data hubungan. Dipilih kerana integriti rekod undang-undang dikuatkuasakan di pangkalan data (RLS + pencetus), bukan di pelayar — baris bertandatangan kekal selamanya.

3.5 Supabase (backend platform — client v2.112.0)

What: a managed backend built on PostgreSQL, bundling the database, authentication, file storage, and auto-generated APIs.

Why chosen: a small team needed a production backend without operating servers. Supabase provides Auth (individual officer logins), Storage (signature images), and a Postgres API in one managed service — with RLS as the single, database-level security model across all of them. It removes the need for a custom application server entirely: the browser talks to Supabase directly, and every row is gated by RLS evaluated as the calling user. This keeps operational burden low (fitting the single-maintainer reality) while keeping integrity high.

Supabase: platform backend terurus atas PostgreSQL (pangkalan data + log masuk + storan). Dipilih supaya pasukan kecil mendapat backend produksi tanpa mengurus pelayan, dengan keselamatan RLS satu lapisan.

3.6 Cloudflare (hosting & CDN — Cloudflare Pages)

What: a global content-delivery network and static-site host (Cloudflare Pages) with edge caching and security headers.

Why chosen: the app is a static bundle, so a CDN is the correct, cheapest, and fastest host — pages load quickly on weak field connections from an edge near the user. Cloudflare Pages builds the repository directly and serves it, applies the security headers (`_headers`: CSP, HSTS, `geolocation=(self)` for "Guna Lokasi Saya"), and — importantly — the rollback is a one-minute DNS switch back to the previous Pages project. Hashed, immutable asset caching plus a no-cache HTML index prevents stale deploys.

Cloudflare: rangkaian penghantaran kandungan (CDN) & hos laman statik. Dipilih kerana pantas di talian lemah, murah, menguatkuasakan pengepala keselamatan, dan membolehkan rollback dalam satu minit.

3.7 Supporting libraries / Pustaka sokongan

Library	Version	Role
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Leaflet	1.9.4	Interactive map rendering (OpenStreetMap tiles). Self-hosted.
Leaflet.markercluster	1.5.3	Clusters hydrant pins at low zoom. Self-hosted.
Node.js	≥20 (CI: 22)	Build & test tooling only — never shipped to the browser.

All third-party libraries are **self-hosted** (no CDN, no SRI needed): a CDN script would run with full access to the signed-in session and every record card. Self-hosting lets the CSP drop to `script-src 'self'`.

3.8 Rationale summary matrix / Ringkasan justifikasi

Layer	Technology	Primary reason chosen
Build	Vite 8.2.1	Strict-CSP-friendly single artifact; fast release gate.
UI	Vue 3.5.41	Testable components; runtime-only (no eval).
State	Pinia 3.0.3	Shared state + pure/testable logic split.
Data	PostgreSQL 15	Integrity in the DB (RLS + triggers); scales to national volume.
Backend	Supabase 2.112	Managed DB+Auth+Storage; no server to operate.
Hosting	Cloudflare Pages	Fast CDN on weak signal; 1-minute rollback; security headers.
Map	Leaflet 1.9.4	Proven, lightweight, self-hostable; no WebGL rewrite needed.

4. Specific Requirements / Keperluan Khusus

4.1 External Interface Requirements / Antara Muka Luaran

4.1.1 User interfaces

- Mobile-first, Bahasa Malaysia UI; large touch targets; no zoom-on-focus; no horizontal scroll; high contrast for sunlight.
- Header tabs: **Peta Pili** (map), **Dashboard**, **Profil**; hamburger navigation below 640px.

4.1.2 Software / communications interfaces

- HTTPS to the Supabase REST/Realtime/Storage endpoints; JWT auth; RLS enforced server-side.
- Map tiles fetched from OpenStreetMap; geolocation API for "Guna Lokasi Saya".

4.2 Functional Requirements / Keperluan Berfungsi

4.2.1 Authentication & Roles / Pengesahan & Peranan

ID	Requirement	Priority
FR-AUTH-1	Users authenticate with an individual email/password login via Supabase Auth.	High
FR-AUTH-2	Any signed-in user may read all data; only <code>admin</code> may write, enforced by RLS.	High
FR-AUTH-3	New accounts default to <code>viewer</code> ; role changes are made by an administrator.	High
FR-AUTH-4	Writes are scoped to the officer's own district; reads remain global (mutual aid).	High

4.2.2 Hydrant Map / Peta Pili

ID	Requirement	Priority
FR-MAP-1	Display all hydrants as clustered pins with status colour (Awam/Swasta) and last-inspection date badge.	High
FR-MAP-2	Filter by status (Awam/Swasta), inspection status, and zone; filters stack with AND.	High
FR-MAP-3	Place-name search; tapping a pin opens a detail modal with coordinates, Directions, and last-inspected.	High
FR-MAP-4	Admins may add a hydrant, validating latitude/longitude and label; "Guna Lokasi Saya" uses device GPS with accuracy feedback.	Med
FR-MAP-5	An amber "!" marks any pin with unsynced local work.	High

4.2.3 Kad Rekod (Inspection Record Card) / Kad Rekod Pemeriksaan

ID	Requirement	Priority
FR-KAD-1	Provide a two-page record card with four sections (Kerosakan, Pemantauan, Pengujian, Kompaun) at fixed per-page row capacities.	High

FR-KAD-2	Automatically create a new card when the last row of any section is completed, on save.	High
FR-KAD-3	A signed row is permanent: never editable, clearable, or deletable, by anyone (RLS + trigger + client).	High
FR-KAD-4	Card numbering is permanent and chronological; newest card first on screen, oldest first in print.	High
FR-KAD-5	Saving the card is the master source for a hydrant's Lokasi and last-inspected date.	Med
FR-KAD-6	Viewers see the card read-only (inputs disabled, Print/Close available).	High

4.2.4 Digital Signatures / Tandatangan Digital

ID	Requirement	Priority
FR-SIG-1	Admins may sign a Pengujian row; the signature image is stored in a private bucket and the row is marked signed.	High
FR-SIG-2	An officer may store one reusable profile signature (a stencil) that fills the sign popup.	Med
FR-SIG-3	A filed signature image is never overwritten (bucket allows INSERT only); replacing the profile stencil writes a new object.	High
FR-SIG-4	Signatures render solid and legible on printed A4 without being erased or blocked out.	High

4.2.5 Dashboard / Papan Pemuka

ID	Requirement	Priority
FR-DASH-1	Derive all figures from the Pengujian rows (Diperiksa / Belum di-sign / Belum diperiksa) — the dashboard stores no numbers of its own.	High
FR-DASH-2	Show status via a 3D donut and status rows; every figure reconciles to the underlying records.	High
FR-DASH-3	Support a rolling 6-month period (current + three archived) and per-zone number ranges (Nombor Pili Terkini).	Med
FR-DASH-4	Donut labels and status rows filter the map; scope follows the Awam/Swasta pills.	Low

4.2.6 Jadual (Schedule) & 4.2.7 Offline Sync / Jadual & Segerak Luar Talian

ID	Requirement	Priority
FR-JAD-1	Provide a shared inspection schedule stored in Supabase (admin-writable), ordered latest date first, foldered by period.	Med
FR-OFF-1	The app is fully usable with no signal; a failed save is parked locally and pushed automatically on reconnect.	High
FR-OFF-2	On a cross-device conflict the cloud copy wins, and the officer is shown exactly what they typed to re-enter or discard.	High

FR-OFF-3	Cross-device changes are picked up on foreground/focus/online and a 60-second poll while visible.	Med
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4.3 Non-Functional Requirements / Keperluan Bukan-Berfungsi

ID	Category	Requirement
NFR-SEC-1	Security	Integrity rules enforced in the database (RLS + triggers); signed rows permanent; audit identity from JWT only. <code>script-src 'self'</code> ; all vendor code self-hosted.
NFR-SEC-2	Security	Signature bucket private; permits INSERT only (no UPDATE/DELETE), preserving evidence permanence.
NFR-PERF-1	Performance	Dashboard entry animation holds 60 fps on a mid-range phone; large reads are paged (1,000-row cap handled).
NFR-REL-1	Reliability	No inspection data lost, including with no signal (offline queue + auto-sync).
NFR-Avail-1	Availability	Static CDN hosting; one-minute rollback via DNS to the previous Pages project; nightly backups verified weekly by automated restore.
NFR-USE-1	Usability	Mobile-first, one-handed, glove-friendly, Bahasa Malaysia, high contrast; usable offline.
NFR-MAINT-1	Maintainability	Component + pure-logic architecture; 30 automated suites, ESLint + type-check gates, 89.9% logic coverage; publishing blocked on tests.
NFR-COMP-1	Compliance	Kad Rekod conforms to MS ISO 9001:2015 PS-8 (2 pages, capacities, retention, permanence).
NFR-PORT-1	Portability	Runs on any modern mobile/desktop browser; no install.

5. Future Plans & Roadmap / Pelan Masa Hadapan

The current system is healthy and live at single-district scale. The improvements below are what the project needs before and during expansion. Items 5.1(2)–5.1(9) are the **organisational actions** identified in the PM review — each requires a decision above the maintainer and is a bigger risk to the project than any code.

Nota: Item 5.1(2)–(9) ialah tindakan organisasi — setiap satu memerlukan keputusan di peringkat pengurusan, dan merupakan risiko lebih besar kepada projek berbanding kod. Perlu diselesaikan sebelum daerah kedua.

5.1 Organisational actions (roadmap items 2–9) / Tindakan Organisasi

#	Action / Tindakan	Why it matters	Owner	Timescale
1	Field test on a real phone — DONE ✓	Verified working in Kunak on a real device (2026-08-18).	Maintainer + officer	Done
2	Recruit & train maintainer #2 Latih penyelenggara kedua	Highest project-ending risk. One person holds the "why"; a transfer/illness strands the app. Negligent at ten districts.	JBPM management	Months (approval)
3	Support & escalation model Model sokongan & eskalasi	"Who does an officer call at 6am?" A blocked, unsupported officer produces bad data trusted by all. Define tiers + published contact/hours.	District managers	Weeks (assign)
4	Data-correction procedure Prosedur pembetulan data	Signed rows are permanent by design; a disputed legal record needs a written, above-maintainer procedure before it is needed.	JBPM policy	Weeks (policy)
5	Backup resilience (offsite) Sandaran luar tapak	Backups live only in 90-day GitHub artifacts that vanish with the account; add an offsite copy of dumps + signature images.	Maintainer	Days
6	Deploy window + approver Tetingkap penerbitan + pelulus	Auto-publish on every commit is unsafe across Sabah where someone is always mid-inspection. Add a release window and an approver.	Maintainer + mgmt	Days
7	Budget & procurement (paid tier) Bajet & perolehan	Supabase free tier ends at scale; a lapsed card stops the app for everyone. Create a budget line; also unblocks leaked-password protection (Pro-only).	JBPM finance	Fiscal year
8	Feature authority + admin-grant policy Kuasa keputusan ciri	At many districts, conflicting requests need someone who can say no with backing; and a defined policy for who may grant admin.	JBPM management	Months

9	Training materials (Bahasa Malaysia) Bahan latihan	~200 officers on shift patterns need written guides + video + a per-station trainer, or they use it wrong and produce trusted-but-wrong data.	JBPM training	Before scale
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5.2 Technical roadmap / Pelan Teknikal

5.2.1 District #2 activation (do when a second station is willing)

The district-ready foundation is already built (a `district` column, `can_write()` scoping writes to the officer's own district, reads global). Activation adds the `upper_admin` (district-manager) role tier, an in-app account-provisioning panel with district-scoped `profiles` RLS and self-escalation guards, and an `All > Kunak > Tawau` selector. Run **one** extra district for three months before designing for ten.

Daerah #2: asas sudah dibina (lajur daerah + kawalan tulis mengikut daerah). Pengaktifan menambah peranan Pengurus Daerah, panel pengurusan akaun, dan pemilih daerah. Jalankan satu daerah tambahan selama tiga bulan dahulu.

5.2.2 National-scale map & geography (future)

Scaling toward state and eventually national volume (~100k hydrants) adds a `state` tier above `district`. The map scales **without** a WebGL/MapLibre rewrite via three legs: (a) state/district filter, (b) bounding-box/viewport loading on a spatial index, and (c) server-side count-bubble aggregation when zoomed out. The hard part is governance (roles/tenancy/audit one layer deeper), not the map.

Skala nasional: menambah lapisan negeri. Peta berskala tanpa menulis semula enjin (penapis + muatan mengikut paparan + agregat kiraan). Bahagian sukar ialah tadbir urus peranan, bukan peta.

5.3 Sequencing / Urutan

Golden rule: settle the organisational actions (5.1 items 2–9) **before** onboarding district #2 — that event is exactly what turns "runs on one person's goodwill" from acceptable into negligent. In the words of the roadmap: *one extra district generates answers; ten generate problems.*

Peraturan emas: selesaikan tindakan organisasi (item 2–9) sebelum memulakan daerah kedua. Satu daerah tambahan memberi jawapan; sepuluh daerah memberi masalah.

Appendix A — Data Model (summary) / Model Data

Table	Holds	Key rules
hydrants	The register (label, lat/Ing, status, location, district).	Read all; write scoped by district; unique (district,label).
hydrant_records	Kad Rekod rows (per section/row_index, data, signed).	Signed rows permanent (RLS + trigger); audit from JWT.
jadual_pemeriksaan	Shared inspection schedule.	Admin-writable; district-scoped.
profiles	Accounts, role, district, profile signature path.	Role defaults to viewer; admin-managed.
Storage: signatures	Signature images (private bucket).	INSERT only — no UPDATE/DELETE (evidence permanence).

Appendix B — Glossary / Glosari

See Section 1.3. Additional: **Diperiksa** = inspected & signed; **Belum di-sign** = filled, awaiting signature; **Belum diperiksa** = not inspected this period; **Zon** = zone, derived from the label's leading letter (A/B/C/D/E), never stored.

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